[2CEIT401 – Operating Systems]

Practical: 6

AIM- Demonstration of process management system calls for multiprocessing.

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Aim: Demonstration of process management system calls for multiprocessing.

1. Create a chain of processes where one parent has exactly one child as per sample chain given by the Instructor.

Sequence:1
PROCESS0--> PROCESS 1--> PROCESS2--> PROCESS3--> PROCESS4--> PROCESS 5--> PROCESS6
Display PIDs of all processes.

Code:

```
max_lvl=6
current_lvl=${1:-0}
echo "Current level: $current_lvl"
echo "Parent Process ID: $PPID"
echo "Process ID: $$"
echo "------------------------------"
if [ $current_lvl -lt $max_lvl ]; then
   sh "$0" $((current_lvl + 1))
fi
```

Output:

```
[24012011142@linuxserv prac6]$ sh que_1.sh
Current level: 0
Parent Process ID: 8823
Process ID: 8855
------------------------------
Current level: 1
Parent Process ID: 8855
Process ID: 8856
------------------------------
Current level: 2
Parent Process ID: 8856
Process ID: 8857
------------------------------
Current level: 3
Parent Process ID: 8857
Process ID: 8858
------------------------------
Current level: 4
Parent Process ID: 8858
Process ID: 8859
------------------------------
Current level: 5
Parent Process ID: 8859
Process ID: 8860
------------------------------
Current level: 6
Parent Process ID: 8860
Process ID: 8861
------------------------------
[24012011142@linuxserv prac6]$
```

2. Write a shell script to calculate the loss percentage of an article, given the cost price and the selling price as command line arguments.

Code:

```
parent_pross_name=$(ps -p $PPID -o comm=)

if [ "$parent_pross_name" = "sh" ]; then
   sorted_args=$(echo "$@" | tr ' ' '\n' | sort -r | tr '\n' ' ')
   echo "Child Process:"
   echo "Reversely Sorted Coins are: $sorted_args"
else
   sorted_args=$(echo "$@" | tr ' ' '\n' | sort | tr '\n' ' ')
   echo "Parent Process:"
   echo "Sorted Coins are: $sorted_args"
   sh "$0" "$@"
fi
```

Output:

```
[24012011142@linuxserv prac6]$ sh que_2.sh 1 2 2 3 4 2 3 4 2 1
Parent Process:
Sorted Coins are: 1 1 2 2 2 2 3 3 4 4
Child Process:
Reversely Sorted Coins are: 4 4 3 3 2 2 2 2 1 1
[24012011142@linuxserv prac6]$
```

3. Write a shell script to accept the name of the user and check out if the same has logged in or not.

Code orphan.c:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        fprintf(stderr, "Fork failed\n");
        return 1;
    }
    else if (pid == 0) {
        // Child process
        sleep(3);
        printf("Child Process ID: %d, New Parent Process ID: %d\n", getpid(), getppid());
    }
    else {
        // Parent process
        printf("Parent Process ID: %d, Child Process ID: %d\n", getpid(), pid);
        exit(0);
    }

    return 0;
}
```

Output:

```
[24012011142@linuxserv prac6]$ ./orphan
Parent Process ID: 8941, Child Process ID: 8942
[24012011142@linuxserv prac6]$ Child Process ID: 8942, New Parent Process ID: 1
```

Code zombie.c:

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
```

```
int main() {
    pid_t pid = fork();

    if (pid > 0) {
        // Parent process
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        sleep(2);

        printf("\nChecking process table (look for Z state):\n");
        system("ps -l");

        sleep(10);
    }
    else if (pid == 0) {
        // Child process
        printf("Child PID: %d exiting...\n", getpid());
        exit(0);
    }
    else {
        perror("Fork failed");
    }

    return 0;
}
```

Output:

```
[24012011142@linuxserv prac6]$ ./zombie
Child PID: 8951 exiting...
Parent PID: 8950
Child PID: 8951

Checking process table (look for Z state):
F S   UID   PID  PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
4 S  8906  8823  8822  0  75   0 -   634 wait   pts/16   00:00:00 bash
0 S  8906  8950  8823  0  75   0 -   371 wait   pts/16   00:00:00 zombie
1 Z  8906  8951  8950  0  77   0 -     0 exit   pts/16   00:00:00 zombie <defunct>
0 R  8906  8952  8950  0  77   0 -   524 -      pts/16   00:00:00 ps
[24012011142@linuxserv prac6]$
```